

Uber Ride Data Analysis

Personal ride history · January – December 2016

413

RIDES ANALYZED

9.1 mi

AVERAGE DISTANCE

44%

MISSING PURPOSE

174 mi

LONGEST TRIP

Executive Summary

This Uber ride log tells a clear story: it's a year of **work travel**. After cleaning the raw file — 1,156 logged rides trimmed down to 413 usable records once bad dates and missing data were removed — the pattern that emerges is short, frequent, weekday trips, mostly in the afternoon and evening, between a handful of familiar locations. The biggest gap in the data is *why* each ride happened: almost half of all entries never got a purpose logged.

Key Insights

CATEGORY

This is business travel, not personal use

The large majority of rides fall under the Business category.

DATA QUALITY

Purpose is the weakest field

About 44% of rides (503 of 1,156) have no recorded purpose. Where it was logged, meetings, errands, and meals show up most often.

TIMING

Weekdays drive demand, afternoons peak

Rides taper off sharply on weekends. Early mornings see the fewest trips; afternoons and evenings are busiest.

DISTANCE

Most trips are short

Median distance is 5.7 miles; the 9.1-mile average is pulled up by a handful of long outlier trips (one as far as 174 miles).

LOCATIONS

A few places account for most trips

Pickups and drop-offs repeat heavily around a small set of locations, suggesting regular routes rather than one-off travel.

What This Means

Taken together, this looks less like scattered personal use and more like a **predictable weekday travel routine** — the kind of pattern that's easy to plan around, budget for, and improve with better data collection. The recommendations on the next page focus on the two things that would make this dataset most useful going forward: closing the purpose-tracking gap, and connecting ride patterns to actual cost.

Recommendations

1. Make "purpose" a required field

It's the single biggest gap in the data — fixing it at the point of booking would unlock much more analysis down the line.

2. Investigate why purpose is missing so often

Check whether it clusters around a specific time, place, or category before assuming it's random.

3. Use the weekday/afternoon pattern for planning

If this reflects a recurring commute or client-visit routine, it's useful for budgeting, scheduling, or negotiating better corporate rates.

4. Treat long outlier trips separately

Rides far beyond the typical 5–10 mile range (likely airport runs or similar) probably deserve their own budget line rather than being averaged into everyday trips.

5. Add fare data if it's available

Pairing distance and timing with actual cost turns this into a spend analysis, not just a usage one — much more useful for budgeting decisions.

The Data, Briefly

The raw file had 1,156 rows and 7 columns (start/end time, category, start/stop location, miles, purpose). Cleaning involved dropping an invalid "Totals" row, converting text dates into real datetime values, filling missing purposes with a placeholder, and removing what remained broken — leaving 413 solid rows. From there, extra columns were built for hour, date, time-of-day bucket, month, and weekday — which is what made the patterns above visible.

A Note for Anyone Reusing the Notebook

One chart references a plotting grid that was never created and will error as written (a working version exists later in the notebook). A couple of chart calls also use an outdated Seaborn function and syntax that newer library versions will flag. The file path is hardcoded to a local folder and will need updating to run elsewhere.